

Quantification: 543_z13

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PRE-VALIDATION: no number here has been checked against manual (ImageJ) counts.
PIXEL UNITS: lengths px, areas px^2, densities px^-1. The images carry no micron calibration.
Statistics are over interior lacunae only (those not touching the frame edge).

Per image

measure	value
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lacuna count	11
interior lacuna count	11
frame-edge lacuna count	0
gap bridges added (detection)	19

Interior lacunae only: mean, median and sample SD (n = 11)

measure	kind	mean	median	sd	units
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area_px2	lacuna shape	1862	1736	823.012	px^2
major_axis_length_px	lacuna shape	78.3615	74.0218	22.3719	px
minor_axis_length_px	lacuna shape	30.8137	31.6227	6.9101	px
aspect_ratio	lacuna shape	2.5873	2.6086	0.6752	unitless
eccentricity	lacuna shape	0.9006	0.9236	0.0676	unitless
solidity	lacuna shape	0.9037	0.9231	0.0388	unitless
roots_count	headline	6.7273	7	2.8316	count
ring_length_r30_px	headline	302.909	310	50.1786	px
ring_length_r60_px	reported	961.364	944	107.799	px
owned_length_px	ownership-dependent	1713.99	1167.87	1503.69	px
edge_count	ownership-dependent	63.5455	43	56.0185	count
mean_edge_length_px	ownership-dependent	26.8926	27.1599	4.8368	px
perimeter_px	normalised	207.479	205.037	54.0901	px
ring_area_r30_px2	normalised	8447.82	8219	1466.9	px^2
ring_area_r60_px2	normalised	21999.7	21794	3275.21	px^2
in_frame_fraction_r30	normalised	1	1	0	unitless
in_frame_fraction_r60	normalised	1	1	0	unitless
ring_density_r30	normalised	0.0362407	0.0361281	0.00540695	px^-1
ring_density_r60	normalised	0.0440284	0.0435812	0.00348523	px^-1
roots_per_100px_perimeter	normalised	3.2312	3.5765	0.9956	per 100 px
ring_attached_length_r30_px	network v2	204.636	192	75.6482	px
ring_attached_length_r60_px	network v2	498.364	412	253.353	px
ring_length_w_r30_px	network v2	341.927	347.605	60.217	px
ring_length_w_r60_px	network v2	1087.24	1066.96	125.671	px
sholl_crossings_r10	network v2	8.0909	8	2.7002	count
sholl_crossings_r20	network v2	11.7273	13	2.9014	count
sholl_crossings_r30	network v2	14.2727	14	2.149	count
ring_width_mean_r30_px	width	4.9871	5.0013	0.4355	px

Per field

measure	value	units
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analysed_area_px2	1.02809e+06	px^2
total_skeleton_length_px	45033	px
canalicular_length_density_per_px	0.0438024	px^-1
junction_count	643	count
junction_density_per_px2	0.00062543	px^-2
lacuna_count	11	count
lacunae_per_px2	1.07e-05	px^-2

skeleton_component_count	659	count
mean_component_length_px	68.3354	px
median_component_length_px	17	px
field_length_density_w_per_px	0.0495823	px^-1
field_density_without_flagged_per_px	0.0438024	px^-1
field_density_in_roi_per_px		px^-1

Canalicular width, the new measure

measure	value	units
width_mean_px	5.7808	px
width_median_px	6	px
width_p10_px	2.8284	px
width_p90_px	8	px
width_pixels_used	45013	count

Canalicular width: 2 x the Euclidean distance transform of the canalicular mask at each skeleton pixel. Measured only on skeleton pixels with real signal under them: bridged pixels, the lacuna buffer and the dilated vascular mask are excluded. The threads are about 2 to 6 px wide, so the width is close to the pixel grid and quantised in steps of about 1 px, it moves with the threshold (a lower cut widens every thread), and it has not been validated against any manual measurement.

Lacuna shape

lacuna_id	area_px2	major_axis_length_px	minor_axis_length_px	aspect_ratio	eccentricity	solidity	orientation_rad	centroid_row_px	centroid_col_px	on_border
1	2019	79.1535	33.2822	2.3783	0.9073	0.9417	0.4378	97.9678	270.286	False
2	839	61.6665	17.863	3.4522	0.9571	0.906	0.068	131.349	828.439	False
3	1736	73.3356	33.3815	2.1969	0.8904	0.8374	0.2312	218.82	919.248	False
4	1138	46.8317	31.6227	1.481	0.7376	0.9252	0.396	263.104	340.498	False
5	2267	87.9083	33.2852	2.6411	0.9255	0.9473	0.0042	459.621	481.093	False
6	860	45.4745	26.4762	1.7176	0.813	0.854	0.07	475.221	748.086	False
7	2150	103.918	27.5311	3.7745	0.9643	0.9231	0.2711	546.673	873.734	False
8	3069	109.359	37.084	2.9489	0.9407	0.9099	0.049	642.67	188.33	False
9	1616	74.0218	28.3763	2.6086	0.9236	0.9245	-0.0477	731.536	630.238	False
10	1438	72.5092	25.5367	2.8394	0.9359	0.9236	0.0953	793.137	478.379	False
11	3350	107.799	44.5116	2.4218	0.9108	0.8475	-0.088	854.917	288.104	False

Network per lacuna

lacuna_id	roots_count	ring_length_r30_px	ring_length_r60_px	owned_length_px	edge_count	mean_edge_length_px	ring_width_mean_r30_px
1	8	331	1011	4094.95	153	26.7644	4.8195
2	7	310	883	1843.99	65	28.3691	5.0013
3	9	343	876	915.009	33	27.7276	4.6128
4	3	226	828	515.688	14	36.8349	5.2431
5	5	326	1067	4189.19	150	27.928	5.4269
6	4	287	861	284.936	16	17.8085	4.8186
7	4	308	1041	167.581	8	20.9476	4.347
8	10	317	955	1859.3	69	26.9463	5.3506
9	7	254	944	1167.87	43	27.1599	5.0347
10	5	234	918	511.961	17	30.1153	4.4422
11	12	396	1191	3303.37	131	25.2166	5.7617

Normalised per lacuna

lacuna_id	perimeter_px	ring_area_r30_px2	ring_area_r60_px2	in_frame_fraction_r30	in_frame_fraction_r60	ring_density_r30	ring_density_r60	roots_per_100px_perimeter
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1	205.037	8499	22530	1	1	0.0389458	0.0448735	3.9017
2	151.297	6982	17707	1	1	0.0443999	0.0498673	4.6267
3	205.037	8219	20045	1	1	0.0417326	0.0437017	4.3895
4	139.681	6734	18999	1	1	0.033561	0.0435812	2.1477
5	223.823	8900	23453	1	1	0.0366292	0.0454953	2.2339
6	141.338	6516	17817	1	1	0.0440454	0.0483246	2.8301
7	254.936	9810	24446	1	1	0.0313965	0.0425837	1.569
8	278.593	10387	26403	1	1	0.0305189	0.0361701	3.5895
9	195.723	8083	21794	1	1	0.031424	0.0433147	3.5765
10	184.995	7835	21305	1	1	0.029866	0.0430885	2.7028
11	301.806	10961	27498	1	1	0.0361281	0.0433122	3.9761

Network v2 per lacuna

lacuna_id	ring_attached_length_r30_px	ring_attached_length_r60_px	ring_length_w_r30_px	ring_length_w_r60_px	sholl_crossings_r10	sholl_crossings_r20	sholl_crossings_r30
1	278	651	391.789	1167.58	11	13	11
2	192	412	335.451	979.11	7	13	13
3	218	388	388.848	986.963	10	15	12
4	120	286	259.037	938.122	3	7	15
5	161	696	364.907	1191.41	7	12	18
6	189	271	327.078	977.436	6	11	15
7	139	252	347.605	1176.64	8	15	13
8	282	748	362.877	1095.16	11	13	17
9	220	664	282.894	1066.96	8	9	13
10	102	166	250.823	1024.34	6	7	14
11	350	948	449.889	1355.87	12	14	16

543_z13: distributions, pre-validation, px

